Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

</tool_call> reasoning process here
The de Broglie wavelength Î» of a particle is given by:

Î» = h / p

where:
- h = Planck's constant = 6.626 ÃĹ 10^-34 J Â·s = 6.626 ÃĹ 10^-27 erg Â·s
- p = momentum of the particle

For a non-relativistic electron, kinetic energy K = (1/2) mv Â², and momentum p = âĪļ(2mK)

We can substitute to get:

Î» = h / âĪļ(2mK)

We need to calculate Î» for K = 1 eV = 1.602 ÃĹ 10^-19 J

The mass of an electron m = 9.109 ÃĹ 10^-31 kg

Now calculate p:

p = âĪļ(2 ÃĹ 9.109 ÃĹ 10^-31 ÃĹ 1.602 ÃĹ 10^-19)
= âĪļ(2.918 ÃĹ 10^-50)
âīĪ 5.402 ÃĹ 10^-25 kg Â·m/s

Now calculate Î»:

Î» = (6.626 ÃĹ 10^-34) / (5.402 ÃĹ 10^-25)
âīĪ 1.227 ÃĹ 10^-9 m

Now convert to cm:
1 m = 100 cm âĨĴ 1.227 ÃĹ 10^-9 m = 1.227 ÃĹ 10^-7 cm

Now check the options in cm:

A. 2.48 ÃĹ 10^-8 cm âĨĴ too small
B. 5.92 ÃĹ 10^7 cm/sec âĨĴ not a wavelength
C. 8.66 ÃĹ 10^-6 cm âĨĴ too large
D. 3.14 ÃĹ 10^-5 cm âĨĴ too large
E. 9.81 ÃĹ 10^-4 cm âĨĴ too large
F. 5.27 ÃĹ 10^-9 m âĨĴ is 5.27 ÃĹ 10^-11 cm âĨĴ not matching
G. 1.23 ÃĹ 10^-7 cm âĨĴ matches 1.227 ÃĹ 10^-7 cm
H. 4.14 ÃĹ 10^-10 m âĨĴ 4.14 ÃĹ 10^-12 cm âĨĴ no
I. 6.625 ÃĹ 10^-27 erg sec âĨĴ not a wavelength
J. 0.512 ÃĹ 10^6 eV âĨĴ energy, not wavelength

So, the correct value is approximately 1.23 ÃĹ 10^-7 cm

<answer> G </answer>